

ArtAdmin - Erfassung kulturhistorischer Objekte

DIPLOMA THESIS

submitted for the

Reife- und Diplomprüfung

at the

Department of Informatik

Submitted by:

Jonas Hinterdorfer
Simon Wögerbauer

Supervisor:

DI David Klewein

Project Partner:

Augustiner-Chorherrenstift St. Florian

Leonding, April 4, 2025

I hereby declare that I have composed the presented paper independently and on my own, without any other resources than the ones indicated. All thoughts taken directly or indirectly from external sources are properly denoted as such.

This paper has neither been previously submitted to another authority nor has it been published yet.

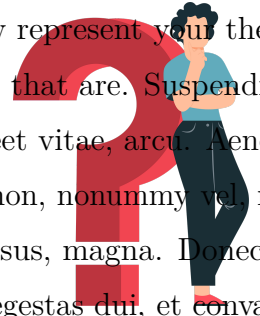
The presented paper is identical to the electronically transmitted document.

Leonding, April 4, 2025

Jonas Hinterdorfer & Simon Wögerbauer

Abstract

Brief summary of our amazing work. In English. This is the only time we have to include a picture within the text. The picture should somehow represent your thesis. This is untypical for scientific work but required by the powers that are. Suspendisse vel felis. Ut lorem lorem, interdum eu, tincidunt sit amet, laoreet vitae, arcu. Aenean faucibus pede eu ante. Praesent enim elit, rutrum at, molestie non, nonummy vel, nisl. Ut lectus eros, malesuada sit amet, fermentum eu, sodales cursus, magna. Donec eu purus. Quisque vehicula, urna sed ultricies auctor, pede lorem egestas dui, et convallis elit erat sed nulla. Donec luctus. Curabitur et nunc. Aliquam dolor odio, commodo pretium, ultricies non, pharetra in, velit. Integer arcu est, nonummy in, fermentum faucibus, egestas vel, odio.



Zusammenfassung

Zusammenfassung unserer genialen Arbeit. Auf Deutsch. Das ist das einzige Mal, dass eine Grafik in den Textfluss eingebunden wird. Die gewählte Grafik soll irgendwie eure Arbeit repräsentieren. Das ist ungewöhnlich für eine wissenschaftliche Arbeit, aber eine Anforderung der Obrigkeit. *Bitte auf keinen Fall mit der Zusammenfassung verwechseln, die den Abschluss der Arbeit bildet!* Suspendisse vel felis. Ut lorem lorem, interdum eu, tincidunt sit amet, laoreet vitae, arcu. Aenean faucibus pede eu ante. Praesent enim elit, rutrum at, molestie non, nonummy vel, nisl. Ut lectus eros, malesuada sit amet, fermentum eu, sodales cursus, magna. Donec eu purus. Quisque vehicula, urna sed ultricies auctor, pede lorem egestas dui, et convallis elit erat sed nulla. Donec luctus. Curabitur et nunc. Aliquam dolor odio, commodo pretium, ultricies non, pharetra in, velit. Integer arcu est, nonummy in, fermentum faucibus, egestas vel, odio.



Contents

1	Introduction	1
1.1	Project Context and Motivation	1
1.2	Initial Situation and Problem Statement	1
1.3	Objectives and Scope	1
2	Current Situation and Related Work	2
2.1	Institutional and Practical Context	2
2.2	Current Problems and Requirements Derived from Practice	2
2.3	Provenance and Collection-Data Standards	2
2.4	Collection Management Systems and Object Identification	4
2.5	Approaches to 3D Measurement	4
3	Proposed Solution	5
3.1	Solution Overview and Requirements	5
3.2	User Roles and Use Cases	5
3.3	Decision Methodology and Data Model	5
3.4	Technical Architecture Decisions	5
3.5	End-to-End Workflows	5
3.6	Security and Integrity Concept	5
4	Technical Description	6
4.1	System Architecture and Frontend	7
4.2	Backend and Data Model	7
4.3	Data Storage and Search	7
4.4	Image Similarity Search	7
4.5	Depth Camera and 3D Measurement	7
4.6	Authentication, Authorization and Auditability	7
4.7	Deployment and Operations	7
4.8	Testing Concept and Traceability	7

5	Evaluation	8
5.1	Evaluation Methodology and Development Process	8
5.2	Provenance Data Model Evaluation	8
5.3	Depth-Camera Measurement Evaluation	8
5.4	Image Similarity and Duplicate Detection Evaluation	8
5.5	System Functionality and Usability Evaluation	8
5.6	Performance and Operational Evaluation	8
5.7	Lessons Learned and Future Improvements	8
5.8	Limitations and Threats to Validity	8
6	Summary	9
	Bibliography	VI
	List of Figures	VII
	List of Tables	VIII
	List of Listings	IX
	Appendix	X

1 Introduction

1.1 Project Context and Motivation

1.2 Initial Situation and Problem Statement

1.3 Objectives and Scope

2 Current Situation and Related Work

2.1 Institutional and Practical Context

2.2 Current Problems and Requirements Derived from Practice

2.3 Provenance and Collection-Data Standards

One of the main requirements of this system is the import and export of information in the form of an established format. This application stores not only physical properties of object but also only digital ones such as a history of the specific object like the creation date or when the data entry was created.

After analyzing existing standards it became clear that no single standard covers all requirements of this project. For that reason the system combined selected componets of several standards. The W3C Prov data model is used to describe the origin and history of the data. Furthermore LIDO and Europeana Data Model is added to store cultural-heritage metadata.

The project specific data schema does not replace these standards. Instead it combines the core features of these and extends them where necessary.

2.3.1 W3C PROV and General Metadata Models

The W3C PROV familiy defines a model for describing the origin and history of any information but it does not define all metadata required for a collection object. It just stores the provenance.

Provenance is information about entities, activities, and people involved in producing a piece of data or thing, which can be used to form assessments about its quality, reliability or trustworthiness [1]

The W3C PROV family consist of multiple documents in this system only used the data model is partially used.

The three main concepts of PROV-DM are:

Entity An entity is a physical, digital, conceptual, or other kind of thing with some fixed aspects; entities may be real or imaginary.[2]

Activity An activity is something that occurs over a period of time and acts upon or with entities; it may include consuming, processing, transforming, modifying, relocating, using, or generating entities.[2]

Agent An agent is something that bears some form of responsibility for an activity taking place, for the existence of an entity, or for another agent's activity.[2]

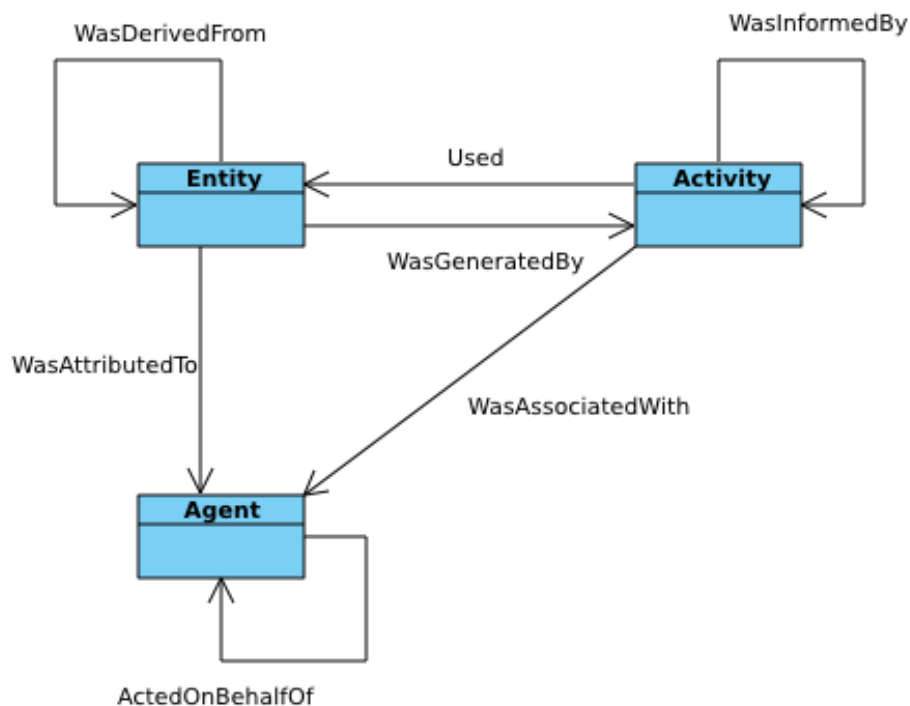


Figure 1: Core concepts of the W3C PROV data model. Source: [2]

2.3.2 CIDOC CRM and Museum-Specific Standards

2.3.3 Operational and Linked-Data Standards

2.4 Collection Management Systems and Object Identification

2.4.1 Existing Collection-Management Solutions

2.4.2 Approaches to Object Identification

2.5 Approaches to 3D Measurement

2.5.1 Traditional and Image-Based Methods

2.5.2 Depth-Camera Technologies and Project Hardware

3 Proposed Solution

3.1 Solution Overview and Requirements

3.2 User Roles and Use Cases

3.3 Decision Methodology and Data Model

3.4 Technical Architecture Decisions

3.4.1 Measurement and Similarity Approaches

3.4.2 Storage and Deployment Architecture

3.5 End-to-End Workflows

3.5.1 Object Registration Workflow

3.5.2 Similarity Search Workflow

3.6 Security and Integrity Concept

4 Technical Description

4.1 System Architecture and Frontend

4.1.1 Overall Architecture

4.1.2 Frontend Implementation

4.2 Backend and Data Model

4.2.1 Backend Architecture

4.2.2 Object and Provenance Data Model

4.2.3 ArtPilot Architecture

4.3 Data Storage and Search

4.3.1 Database and Media Management

4.3.2 Full-Text Search and Filtering

4.4 Image Similarity Search

4.4.1 Processing Pipeline

4.4.2 Similarity Computation and Robustness

4.5 Depth Camera and 3D Measurement

4.5.1 Hardware, Software and Measurement Principle

4.5.2 Data Quality and Measurement Uncertainty

4.5.3 Measurement versus Scanning and Limitations

5 Evaluation

5.1 Evaluation Methodology and Development Process

5.2 Provenance Data Model Evaluation

5.3 Depth-Camera Measurement Evaluation

5.3.1 Experimental Setup and Metrics

5.3.2 Results and Interpretation

5.4 Image Similarity and Duplicate Detection Evaluation

5.4.1 Dataset and Metrics

5.4.2 Threshold Evaluation and Practical Usability

5.5 System Functionality and Usability Evaluation

5.6 Performance and Operational Evaluation

5.7 Lessons Learned and Future Improvements

5.8 Limitations and Threats to Validity

6 Summary

Aufzählungen:

- Itemize Level 1
 - Itemize Level 2
 - Itemize Level 3 (vermeiden)
- 1. Enumerate Level 1
 - a. Enumerate Level 2
 - i. Enumerate Level 3 (vermeiden)

Desc Level 1

Desc Level 2 (vermeiden)

Desc Level 3 (vermeiden)

Bibliography

- [1] P. Groth and L. Moreau, “PROV-overview: An overview of the PROV family of documents,” World Wide Web Consortium, W3C Working Group Note, Apr. 2013. [Online]. Available: <https://www.w3.org/TR/prov-overview/>
- [2] L. Moreau and P. Missier, “PROV-DM: The PROV data model,” World Wide Web Consortium, W3C Recommendation, Apr. 2013. [Online]. Available: <https://www.w3.org/TR/2013/REC-prov-dm-20130430/>

List of Figures

1	Core concepts of the W3C PROV data model. Source: [2]	3
---	---	---

List of Tables

List of Listings

Appendix